

StudyHelper – Project Description

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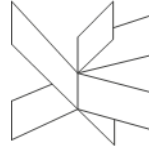
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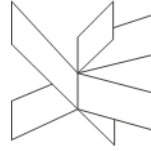
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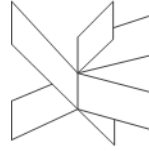
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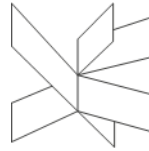
1. Problem Domain

“The physical environment in which the educational process takes place has been the subject of study in recent years, given its direct impact on the well-being and academic performance of students” (Bustamante-Mora et al., 2025, para. 2.) Environmental conditions such as temperature, noise, state-of-mind and sleep quality can have either a positive or negative effect on the ability to study, even when the students have required motivation. This can cause them to shift places, constantly adjust lighting and clothing, etc. This results in a waste of time and energy.

This problem is particularly visible in shared spaces, such as libraries, classrooms and dorms, where the infrastructure is suboptimal for studying and working. For example, many students report a lack of focus in a noisy space due to a lack of soundproofing. This forces them to opt out for noise-cancelling headphones or white-noise machines. Unfortunately, these solutions are not widely available, meaning that many continue to struggle with subpar conditions.

There is a clear need for a system that closes the loop from raw environmental data to cognitive efficiency. By integrating IoT sensors with Machine Learning, we can move beyond simple data logging to predictive modeling.

The stakeholders in this problem are students, who need to maintain focus, motivation and low stress levels; educational institutions, who are the providers of study-spaces; study-space administrators, who manage shared learning environments.



2. Problem statement

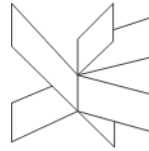
Based on the environmental challenges identified in the problem domain, it is clear that the condition of the environment affects the level of productivity and concentration in students . Therefore, this project addresses the following:

Main problem:

- How can indoor environmental data be monitored and analyzed to understand the relationship between atmospheric conditions and students' focus during study sessions?

Sub-questions:

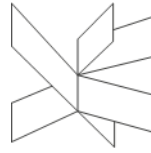
- What are the most significant indoor environmental factors that affect the efficiency of students' studies?
- What factors can be used to represent the students' efficiency of studies and the atmospheric quality?
- How do students react to the changes in the atmospheric quality?
- Is there any relationship between the environmental factors and the students' efficiency of studies?
- How accurately can the future effects be predicted based on the collected data?

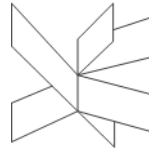


3. Delimitation

This project will be focused on creating a distributed IoT-based Study Environment Monitoring System with the ability to measure conditions of the environment and predict the levels of focus in a study area. In order for the project to be feasible under the scope of a semester project, several delimitations were identified and defined.

1. Geographical Scope
 - The system will be designed and tested within multiple study environments, such as in one campus.
2. System Features
 - The system will include some measurement with different sensors.
 - There will be no inclusion of medical analysis, psychological analysis, and university systems such as LMS and grading.
 - There will be no full automation of the room's devices, such as heating, ventilation, and air conditioning.
3. Actors and Stakeholders
 - There will only be one actor for the system, the student.
 - There will be no inclusion of parents, the university, healthcare professionals, and others.
4. Data and Privacy
 - There will only be the collection of environmental data.
 - If the system includes a microphone, the inclusion of sound level measurement (dB only) will be implemented.
 - There will be no video monitoring.





4. Choice of methods

4.1 Development Methodology

The development of the system will follow the Agile Unified Process (AUP) combined with Scrum as the project management framework (Ambler, 2006). Thanks to AUP, the development process will be structured into the phases Inception, Elaboration, Construction, and Transition. This process supports iterative and incremental development, allowing the team to improve the system architecture and functionality step by step during the project (Ambler, 2006).

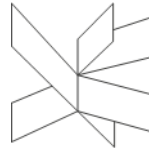
For development coordination, Large-Scale Scrum will be applied (Larman & Vodde, 2016). The team is split into three sub-teams: IoT, Machine Learning, and Front-End, all working within the same sprint cycle. Each team has its own Scrum Master, while a single Product Owner manages the overall product backlog and prioritizes system requirements (Larman & Vodde, 2016).

Sprint durations will depend on the development phase. During the Elaboration phase, the teams will work in two-week sprints, while during the Construction phase the sprint duration will be one week to support faster implementation and integration. Each Scrum team will conduct its own ceremonies and adjust them according to the team's needs.

4.2 Requirements Analysis

The team defines system requirements through user stories and use cases. The team writes user stories according to the SMART principles (Doran, 1981). Each story defines a clear goal and measurable outcome. This structure guides development and supports clear task definition.

The team creates several UML diagrams to support system analysis and design. These include Use Case diagrams, Sequence diagrams, Activity diagrams, a Domain Model, and Class diagrams. These diagrams describe system behaviour and structure.



4.3 Development and DevOps Practices

The project follows DevOps practices to support collaboration and continuous integration between teams (Bass, Weber & Zhu, 2015). The team stores the source code in Git and uses GitHub as the central repository.

Automated tests run during development. These tests check whether new code changes break parts of the system which already worked correctly.

Docker containers will run parts of the system where needed. Containers keep the development and runtime environments consistent.

The backend runs on a public cloud provider which hosts the system services.

4.4 System Implementation

The system includes several main components. The IoT hardware platform uses the ATmega2560 MCU. The team programs the device in C. The device collects environmental data from sensors and sends the data to the backend system.

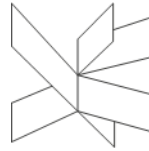
The backend is developed in C#. The backend exposes functionality through a REST API and handles communication between the IoT devices, the database, and the web application.

The system stores data in a PostgreSQL database. The database stores sensor measurements and other system data.

The machine learning component processes the collected sensor data and generates predictions based on the observed patterns. The team develops the model in Python.

The frontend is a responsive web application developed in React. The application displays sensor data, predictions, and historical information to the user.

4.5 Testing



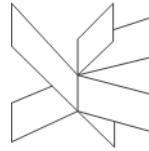
Testing takes place during the entire development process. The team writes unit tests for individual components of the system (Beck, 2003). These tests check whether each component works as expected.

Automated tests also run during development. These tests check whether new code changes break existing functionality.

4.6 Project Management and Communication

Project tasks will be managed using GitHub and Figma. These tools support task tracking, collaboration between teams and design of the user interface.

Communication within the development team will primarily take place through Discord, which will be used for daily coordination, discussions and information sharing between the teams.



5. Time schedule

Number of group members	13 members
ECTS per student	10 ECTS
Expected workload per ECTS per student	28 hours
Sprint number of days between 16th of March and 4th of May	14 days
Sprint number of days between 4th of May and 28th of May	7 days

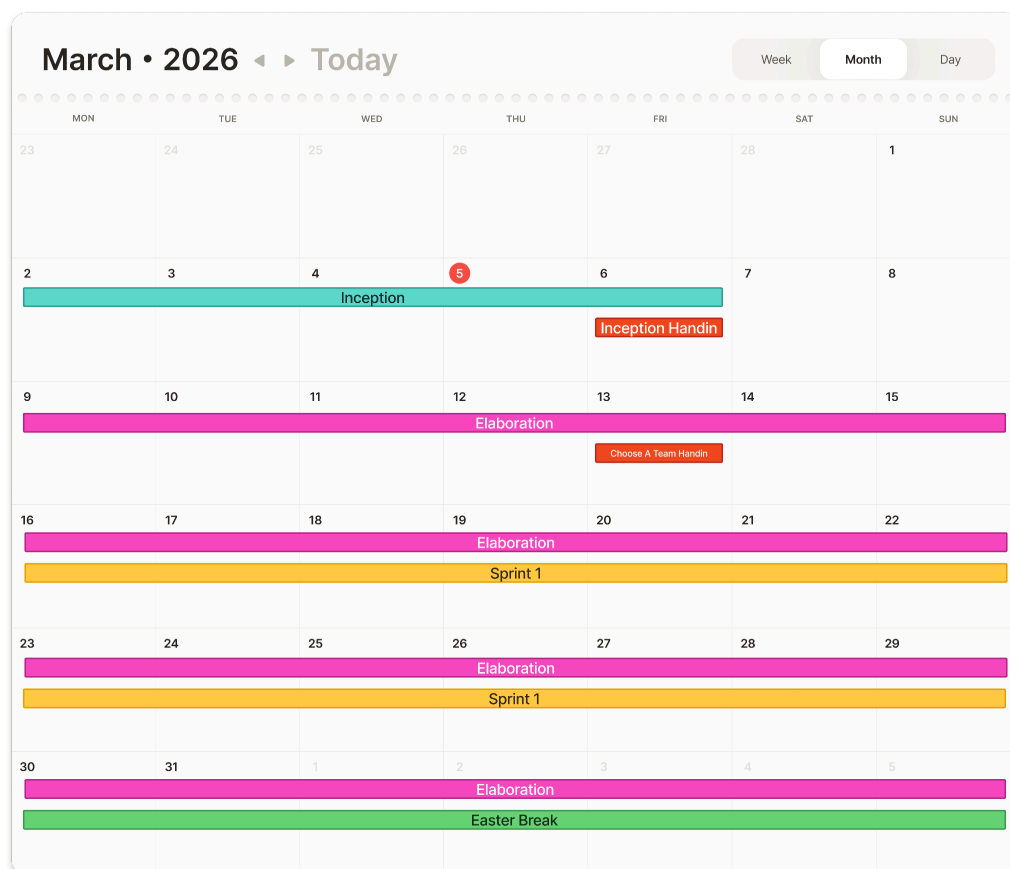


Figure 1 - March Schedule

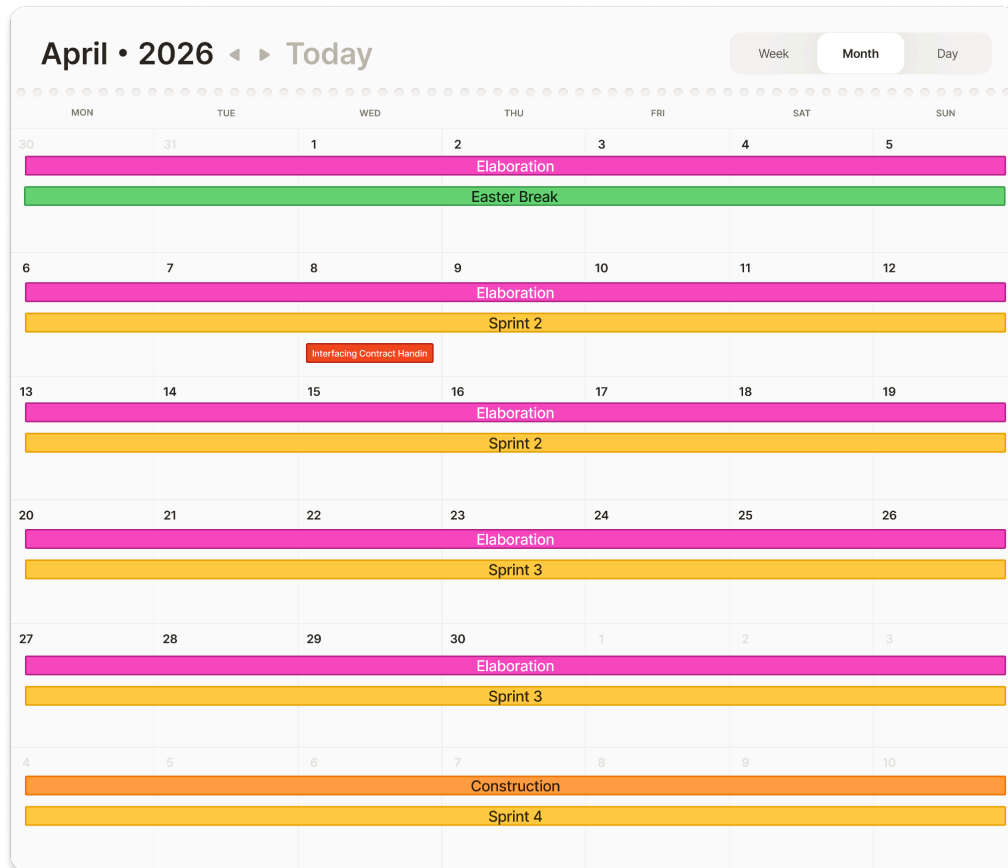
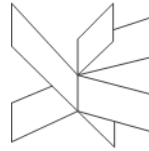


Figure 2 - April Schedule

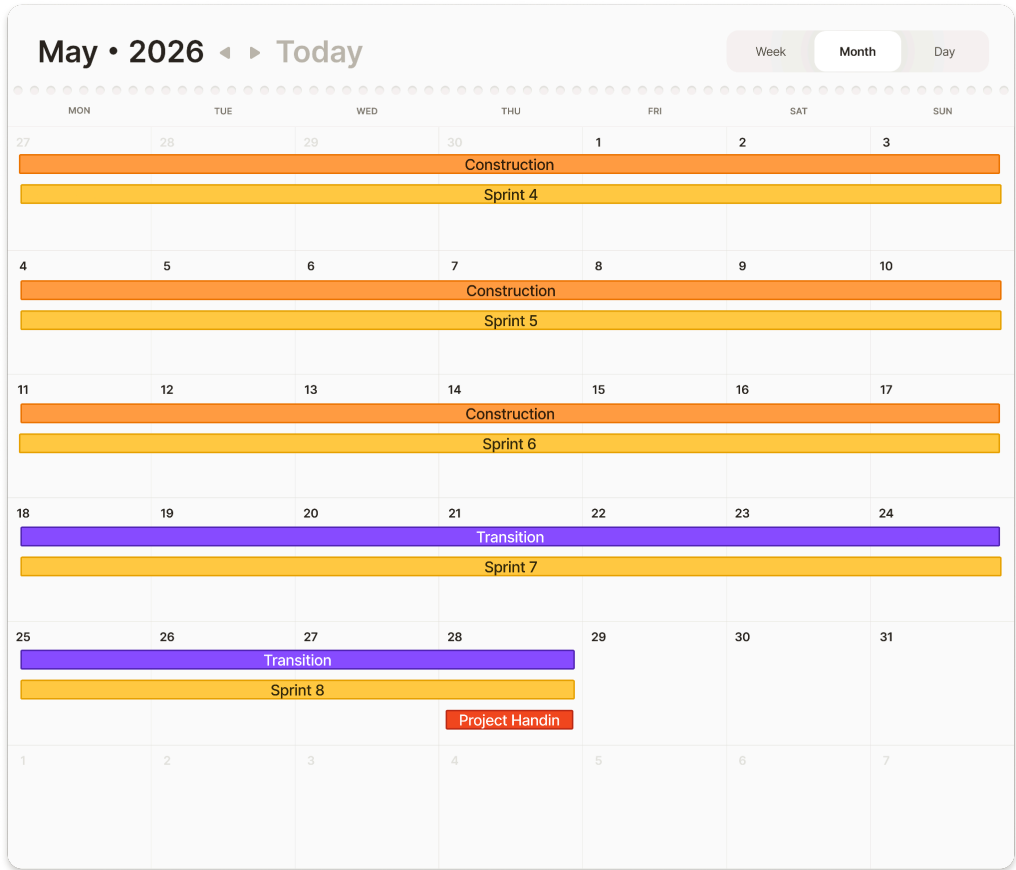
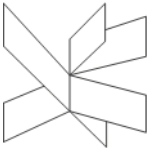
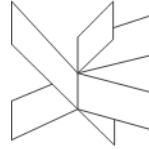


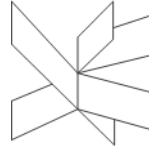
Figure 3 - May Schedule

Figures 1–3 show the detailed schedules for all planned Agile Unified Process development stages (Larman & Vodde, 2016). The figures also include the planned sprint timelines and the corresponding hand-in dates.



6. Risk assessment

Risk description	Likelihood (0 - 5)	Impact (0 - 5)	Risk Score (Likelihood x Impact)	Preventive Actions
Insufficient dataset for ML model training	4	5	20	Create dummy data
Late integration may occur if each group prioritizes its internal objectives without regularly aligning with other teams	4	5	20	Hold regular meetings to review and define clear integration deadlines
A delay in one team may block other teams from making progress	4	4	16	Set shared milestones and monitor progress regularly to identify and address delays early
Hardware malfunction in IoT team	3	4	12	Perform hardware test as early as possible
Decision-making delays due to large team size	2	3	6	Use majority voting when consensus cannot be reached within a defined discussion period



7. References

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